

Pseudocode

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ALGORITHM BellmanFord(Graph G, sourceNode S)
  INPUT: Graph G (weighted), source node S
  OUTPUT: distance[]

  1. Initialize:
    distance[S] = 0
    distance[other nodes] = INFINITY

  2. FOR i = 1 to |V|-1: // |V| = jumlah vertex
    FOR each edge (u, v) in G:
      IF distance[u] + weight(u,v) < distance[v]:
        distance[v] = distance[u] + weight(u,v)

  3. // Check negative cycle
    FOR each edge (u, v) in G:
      IF distance[u] + weight(u,v) < distance[v]:
        RETURN "Negative cycle detected"

  4. RETURN distance[]
END ALGORITHM

```

distance[S] = 0

distance[other nodes] = INFINITY

(Kompleksitas : $O(V)$)

FOR i = 1 to |V|-1:

FOR each edge (u, v) in G:

(Kompleksitas : $O(V \times E)$)

FOR each edge (u, v) in G:

(Kompleksitas : $O(E)$)

$O(V) + O(VE) + O(E) = O(V \times E)$

Coding

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class BellmanFord:
    def __init__(self):
        self.edges = []
        self.nodes = set()

    def add_edge(self, u, v, weight):
        self.edges.append((u, v, weight))
        self.nodes.add(u)
        self.nodes.add(v)

    def shortest_path(self, start):
        distances = {node: float('inf') for node in self.nodes}
        distances[start] = 0

        for _ in range(len(self.nodes) - 1):
            for u, v, weight in self.edges:
                if distances[u] != float('inf') and distances[u] + weight < distances[v]:
                    distances[v] = distances[u] + weight

        for u, v, weight in self.edges:
            if distances[u] != float('inf') and distances[u] + weight < distances[v]:
                return None, "Negative cycle detected!"

        return distances, None

bellman = BellmanFord()
bellman.add_edge('A', 'B', 4)
bellman.add_edge('A', 'C', 2)
bellman.add_edge('B', 'C', -1)
bellman.add_edge('B', 'D', 5)
bellman.add_edge('C', 'D', 8)

distances, error = bellman.shortest_path('A')
print("\n=== BELLMAN-FORD ===")
if error:
    print(error)

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else:
    print("Jarak dari A:")
    for node, dist in distances.items():
        print(f"  A → {node}: {dist}")
```

```
=== BELLMAN-FORD ===
Jarak dari A:
A → B: 4
A → C: 2
A → A: 0
A → D: 9
```